GATE DA: LINEAR ALGEBRA MASTER QUESTION BANK

Complete 250 Questions Benchmark — Syllabi-Aligned, Actual PYQs, Step-by-Step Solutions & Theory Reference

Document Architecture & Topic Weightage: Matrices & Matrix Operations (Q1–Q30) | Determinants (Q31–Q55) | Rank of Matrix (Q56–Q80) | Systems of Linear Equations (Q81–Q110) | Vector Spaces & Subspaces (Q111–Q135) | Linear Independence & Dependence (Q136–Q160) | Basis & Dimension (Q161–Q185) | Eigenvalues & Eigenvectors (Q186–Q215) | Diagonalization (Q216–Q235) | Orthogonality, Gram-Schmidt & Projections (Q236–Q250).

CHAPTER 1: MATRICES & MATRIX OPERATIONS (Q1 – Q30)

Q1. [GATE DA 2024, Question 2]

MCQ | Moderate

Let A and B be $n \times n$ real non-zero matrices such that $AB = 0$. Which one of the following statements is ALWAYS TRUE?

(A) Rank(A) + Rank(B) ≤ n
(B) Either $A = 0$ or $B = 0$
(C) Rank(A) = Rank(B)
(D) det(A) + det(B) ≠ 0

Q2. [GATE CS 2021, Set 1]

NAT | Easy

Let M be a 3×3 matrix with real entries such that $M^2 = M + 2I$. If the trace of M is 1 and det(M) is −4, the value of trace(M^3) is _____.

Q3. [GATE DA 2024, Question 23]

MCQ | Moderate

Let A be an $n \times n$ invertible real matrix. Consider the block matrix $M = [A, I; I, 0]$. Then M^{-1} is given by:

(A) $[0, I; I, -A]$
(B) $[-A, I; I, 0]$
(C) $[0, I; -I, A]$
(D) $[A^{-1}, 0; 0, -A]$

Q4. [GATE CS 2017]

MCQ | Easy

Let P and Q be $n \times n$ matrices such that $P^2 = P$ and $Q^2 = Q$. If $P + Q = I$, which of the following is correct?

(A) $PQ = QP = 0$
(B) $PQ = I$
(C) $P^T = Q$
(D) P and Q must both be invertible

Q5. [GATE EC 2018]

NAT | Moderate

Let A = [1, 2; 3, 4] and B = [2, 0; 1, 3]. The trace of A^TB is _____.

Q6. [Advanced Conceptual]

MSQ | Difficult

Let $A, B \in \mathbb{R}^{n \times n}$. Which of the following statements is/are ALWAYS TRUE?

(A) Trace(AB) = Trace(BA)
(B) Trace(ABC) = Trace(BAC) for any $C \in \mathbb{R}^{n \times n}$
(C) If A is skew-symmetric and n is odd, det(A) = 0
(D) $A^TA = 0$ implies $A = 0$

Q7. [GATE DA Expected 2027] NAT | Moderate

Let A be a 4×4 nilpotent matrix of index 3 (i.e., $A^3 = 0$ but $A^2 \neq 0$). What is the value of $\det(I + 2A + 3A^2)$?

Q8. [Application-Based] MCQ | Moderate

In machine learning, the Gram matrix G of data matrix $X \in \mathbb{R}^{N \times d}$ is defined as $G = XX^T$. Which property does G ALWAYS satisfy?

(A) Symmetric and Positive Semi-Definite
(B) Strictly Positive Definite
(C) Orthogonal
(D) Skew-Symmetric

Q9. [GATE CS 2014] NAT | Easy

Let A be a 3×3 matrix with real entries such that $\det(A) = 6$. The value of $\det(2A^{-1})$ is _____.

Q10. [GATE EE 2019] NAT | Moderate

If $A = [0, 1; -1, 0]$, then the matrix A^{102} is equal to kI. The value of k is _____.

Q11. [Advanced Conceptual] MSQ | Difficult

Let A be an $n \times n$ real matrix such that $A^k = 0$ for some positive integer k. Which of the following is/are TRUE?

(A) $I - A$ is invertible with inverse $I + A + A^2 + \ldots + A^{k-1}$
(B) All eigenvalues of A are 0
(C) Trace(A) = 0
(D) A must be symmetric

Q12. [Mixed Concept] NAT | Difficult

Let $u = [1, 2, 2]^T$. Let matrix $H = I - 2\,(u\,u^T) / (u^T\,u)$ (Householder reflection matrix). The absolute value of $\det(H)$ is _____.

Q13. [GATE DA Expected 2027] MCQ | Moderate

Let $A, B \in \mathbb{R}^{n \times n}$ such that $AB - BA = I$. Which of the following statements is correct?

(A) Such matrices A and B do not exist
(B) A must be invertible
(C) B must be nilpotent
(D) Both A and B are orthogonal

Q14. [GATE IN 2020] NAT | Easy

If $A = [\cos(\theta), -\sin(\theta); \sin(\theta), \cos(\theta)]$, then the determinant of A^{50} for $\theta = \pi/6$ is _____.

Q15. [GATE ME 2016] NAT | Moderate

For the matrix $A = [1, 3; 0, 1]$, the entry in row 1, column 2 of the matrix A^{25} is _____.

Q16. [GATE CS 2015] MCQ | Easy

If an $n \times n$ matrix satisfies $A^T = -A$, then A is called:

(A) Skew-symmetric
(B) Symmetric
(C) Involutory
(D) Idempotent

Q17. [Advanced Conceptual] NAT | Moderate

Let A be a 3×3 orthogonal matrix with $\det(A) = 1$. If $\text{trace}(A) = 1$, then the angle of rotation (in radians) represented by A in $[0, \pi]$ is π / c. The integer value of c is _____.

Q18. [GATE DA Expected 2027] MSQ | Difficult

Let P be an idempotent matrix ($P^2 = P$). Which of the following is/are ALWAYS TRUE?

(A) The eigenvalues of P can only be 0 or 1
(B) $I - P$ is idempotent
(C) $\text{Trace}(P) = \text{Rank}(P)$
(D) P is always invertible if $P \neq 0$

Q19. [Application-Based] NAT | Moderate

In a Markov chain, the transition matrix is $P = [0.7, 0.3; 0.2, 0.8]$. If the steady-state probability vector is $\pi = [\pi_1, \pi_2]$ such that $\pi P = \pi$ and $\pi_1 + \pi_2 = 1$, the value of π_1 is _____ (round off to 2 decimal places).

Q20. [GATE EC 2021] NAT | Moderate

Let $A = [2, 1; 0, 2]$. Then the trace of e^A is _____ (expressed in terms of e, round off to nearest integer, where $e \approx 2.718$).

Q21. [GATE CS 2018] NAT | Easy

Consider the matrix $A = [1, 1, 1; 1, 1, 1; 1, 1, 1]$. The value of $\text{trace}(A^4)$ is _____.

Q22. [GATE DA Expected 2027] MCQ | Difficult

Let $A, B \in \mathbb{R}^{n \times n}$ such that A is symmetric and B is skew-symmetric. Then the matrix $A\,B\,A$ is:

(A) Symmetric
(B) Skew-symmetric
(C) Orthogonal
(D) Identity

Q23. [Advanced Conceptual] NAT | Moderate

Let A be a 3×3 involutory matrix (i.e. $A^2 = I$). If $\text{trace}(A) = 1$, then the determinant of A is _____.

Q24. [Mixed Concept] MSQ | Difficult

Let A and B be $n \times n$ real symmetric matrices. Which of the following is/are TRUE?

(A) AB is symmetric if and only if $AB = BA$
(B) $A + B$ is always symmetric
(C) $AB + BA$ is always symmetric
(D) $AB - BA$ is always skew-symmetric

Q25. [GATE CS 2012] NAT | Easy

Let $A = [1, 2; 0, 1]$. The inverse of A is $A^{-1} = [1, x; 0, 1]$. The value of x is _____.

Q26. [GATE DA Expected 2027] NAT | Moderate

Let $x = [1, -1, 2]^T$. The trace of the rank-1 outer product matrix $M = x\,x^T$ is _____.

Q27. [Application-Based] MCQ | Moderate

Let K be an adjacency matrix of an undirected simple graph with n vertices. The (i, j)-th entry of K^2 represents:

(A) Number of paths of length 2 between vertex i and vertex j
(B) Shortest distance between i and j
(C) Whether i and j share an edge
(D) Degree of vertex i plus vertex j

Q28. [GATE EE 2017] NAT | Easy

If A is a 2×2 matrix with trace = 4 and det = 3, then trace(A^2) is _____.

Q29. [Advanced Conceptual] MCQ | Difficult

Let A be an $m \times n$ matrix and B be an $n \times m$ matrix. Then:

(A) Non-zero eigenvalues of AB and BA are identical including multiplicities
(B) det(AB) = det(BA) for all m, n
(C) Trace(AB) ≠ Trace(BA) when $m \neq n$
(D) Rank(AB) = Rank(BA) always

Q30. [Mixed Concept] NAT | Difficult

Let J_n be the $n \times n$ all-ones matrix. Let $A = 2I_4 + 3J_4$. The trace of A is _____.

CHAPTER 2: DETERMINANTS (Q31 – Q55)

Q31. [GATE DA 2024, Question 34] NAT | Moderate

Let A be a 3×3 matrix with det(A) = 5. The value of det(adj(adj(A))) is _____.

Q32. [GATE CS 2019] NAT | Easy

Let A be a 4×4 real matrix. If det($3A$) = 162, then the absolute value of det(A) is _____.

Q33. [GATE CE 2020] NAT | Moderate

The determinant of the matrix
[1, 1, 1; 1, 2, 4; 1, 3, 9] is _____.

Q34. [GATE ME 2018] MCQ | Moderate

If A is an $n \times n$ skew-symmetric matrix and n is odd, then det(A) is:

(A) 0
(B) 1
(C) −1
(D) Depends on entries

Q35. [Advanced Conceptual] MSQ | Difficult

For an $n \times n$ invertible matrix A, which of the following identities is/are ALWAYS TRUE?

(A) $\det(\text{adj}(A)) = (\det(A))^{n-1}$
(B) $\text{adj}(\text{adj}(A)) = (\det(A))^{n-2} A$
(C) $\det(A^{-1}) = 1 / \det(A)$
(D) $\det(A + B) = \det(A) + \det(B)$

Q36. [GATE DA Expected 2027] NAT | Difficult

Evaluate the determinant of the 4×4 circulant matrix C whose first row is [1, 2, 3, 4] and each subsequent row is a cyclic right shift of the previous row. $\det(C) =$ _____.

Q37. [Application-Based] NAT | Moderate

In high-dimensional multivariate statistics, the covariance matrix of a 2D Gaussian is Σ = [4, 2; 2, 3]. The determinant of Σ, representing generalized variance, is _____.

Q38. [GATE EC 2016] NAT | Easy

If matrix A = [2, x; 4, 8] is singular, then the value of x is _____.

Q39. [GATE CS 2010] NAT | Moderate

The determinant of the matrix
[1+a, 1, 1; 1, 1+b, 1; 1, 1, 1+c] with $a = b = c = 1$ is _____.

Q40. [Advanced Conceptual] MCQ | Difficult

Let $A, B, C, D \in \mathbb{R}^{n \times n}$ form a block matrix $M = [A, B; C, D]$. If A is invertible and $AC = CA$, then $\det(M)$ equals:

(A) $\det(AD - BC)$
(B) $\det(AD - CB)$
(C) $\det(A)\det(D) - \det(B)\det(C)$
(D) $\det(DA - BC)$

Q41. [GATE DA Expected 2027] NAT | Moderate

Let A be a 5×5 real matrix such that $A^T = -A$. The value of $\det(A^3 + A)$ is _____.

Q42. [GATE CS 2008] NAT | Moderate

Let A be a 3×3 matrix where $a_{ij} = i \times j$ for all $1 \le i, j \le 3$. The determinant of A is _____.

Q43. [GATE EE 2015] NAT | Easy

Let P = [1, 2; 3, 5]. The determinant of P^{-1} is _____.

Q44. [Mixed Concept] MSQ | Difficult

Which of the following operations on a square matrix A PRESERVES the determinant of A?

(A) Transposing the matrix: $\det(A^T) = \det(A)$
(B) Adding a scalar multiple of one row to another row
(C) Multiplying a single row by a non-zero scalar c
(D) Conjugation by an invertible matrix P: $\det(P^{-1}AP) = \det(A)$

Q45. [Application-Based] NAT | Moderate

The Jacobian determinant for coordinate transformation from Cartesian (x, y, z) to spherical (r, θ, ϕ) has the form $r^k \sin(\theta)$. The integer exponent k is _____.

Q46. [GATE IN 2018] NAT | Easy

Determinant of the tridiagonal matrix
[2, 1, 0; 1, 2, 1; 0, 1, 2] is _____.

Q47. [Advanced Conceptual] MCQ | Moderate

Let $u, v \in \mathbb{R}^n$ with $n \geq 2$. Then $\det(I + u\,v^T)$ is equal to:

(A) $1 + v^T u$
(B) $(1 + v^T u)^n$
(C) $u^T v$
(D) 1

Q48. [GATE DA Expected 2027] NAT | Difficult

Let A be a 4×4 matrix such that $\det(A) = -2$. The determinant of matrix $B = 2\,A^{-1} + adj(A)$ equals _____.

Q49. [GATE CE 2014] NAT | Easy

Determinant of [1, 0, 0; 2, 3, 0; 4, 5, 6] is _____.

Q50. [GATE CS 2005] NAT | Moderate

If A is a 3×3 matrix with $\det(A) = 4$, then $\det(\text{adj}(A))$ is _____.

Q51. [Advanced Conceptual] NAT | Difficult

Let A be a 3×3 orthogonal matrix. If $\det(A - I) \neq 0$, then $\det(A)$ must equal _____.

Q52. [Mixed Concept] MSQ | Difficult

Let $A \in \mathbb{R}^{n \times n}$. Which of the following conditions guarantee that $\det(A) = 0$?

(A) The rows of A are linearly dependent
(B) 0 is an eigenvalue of A
(C) The nullity of A is strictly greater than 0
(D) $\text{Trace}(A) = 0$

Q53. [GATE DA Expected 2027] NAT | Moderate

Let $A = [3, 1; 2, 4]$. The determinant of $A^3 - 7A^2 + 10A$ is _____.

Q54. [Application-Based] MCQ | Moderate

In affine geometry, the area of a triangle with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ is given by half the absolute value of the determinant of which matrix?

(A) $[x_1, y_1, 1; x_2, y_2, 1; x_3, y_3, 1]$
(B) $[x_1, y_1, 0; x_2, y_2, 0; x_3, y_3, 1]$
(C) $[x_1, x_2, x_3; y_1, y_2, y_3; 0, 0, 0]$
(D) $[x_1, y_1; x_2, y_2]$

Q55. [GATE ME 2015] NAT | Easy

For a 2×2 matrix with eigenvalues 2 and 5, the determinant is _____.

CHAPTER 3: RANK OF MATRIX (Q56 – Q80)

Q56. [GATE DA 2024, Question 15] NAT | Moderate

The rank of the matrix
$M = [1, 2, 3, 4; 2, 4, 6, 8; 3, 6, 9, 12; 4, 8, 12, 16]$ is _____.

Q57. [GATE CS 2020] NAT | Easy

The rank of the matrix $A = [1, 1, 0; 0, 1, 1; 1, 0, 1]$ is _____.

Q58. [GATE EC 2017] NAT | Moderate

The rank of the matrix
$[1, -1, 0; 0, 1, -1; -1, 0, 1]$ is _____.

Q59. [Advanced Conceptual] MSQ | Difficult

Let $A \in \mathbb{R}^{m \times k}$ and $B \in \mathbb{R}^{k \times n}$. Which of the following inequalities ALWAYS HOLD?

(A) $\text{Rank}(AB) \leq \min(\text{Rank}(A), \text{Rank}(B))$
(B) Sylvester's rank inequality: $\text{Rank}(AB) \geq \text{Rank}(A) + \text{Rank}(B) - k$
(C) $\text{Rank}(A + B) \leq \text{Rank}(A) + \text{Rank}(B)$ (when $m = k = n$)
(D) $\text{Rank}(A^T A) = \text{Rank}(A)$ for any real matrix A

Q60. [GATE DA Expected 2027] NAT | Difficult

Let A be a 5×5 matrix with $\text{Rank}(A) = 3$. What is the rank of $\text{adj}(A)$?

Q61. [Application-Based] MCQ | Moderate

In Principal Component Analysis (PCA) on a centered data matrix $X \in \mathbb{R}^{N \times d}$ where $N = 1000$ and $d = 10$, if the rank of X is 7, the maximum number of non-zero principal component variances is:

(A) 7
(B) 10
(C) 1000
(D) 3

Q62. [GATE CS 2016] NAT | Easy

Rank of the matrix $[1, 2, 3; 2, 4, 6; 0, 0, 0]$ is _____.

Q63. [GATE EE 2021] NAT | Moderate

For what value of k does the matrix
$[1, 2, 1; 2, 5, 3; 3, 7, k]$ have rank 2?

Q64. [Advanced Conceptual] MCQ | Moderate

If A is an $n \times n$ matrix with $\text{Rank}(A) = n - 1$, then $\text{Rank}(\text{adj}(A))$ is:

(A) 1
(B) 0
(C) $n - 1$
(D) n

Q65. [GATE DA Expected 2027] NAT | Moderate

Let $u = [1, 2, 3]^T$ and $v = [4, 5, 6]^T$. Let $A = u\,v^T + v\,u^T$. The rank of A is _____.

Q66. [GATE ME 2019] NAT | Easy

The rank of a 4×4 identity matrix is _____.

Q67. [Mixed Concept] MSQ | Difficult

Let $A \in \mathbb{R}^{m \times n}$. The Rank-Nullity Theorem states that:

(A) Rank(A) + Nullity(A) = n
(B) Dimension of Column Space + Dimension of Null Space = Number of columns
(C) Rank(A) + Nullity(A^T) = m
(D) Rank(A) = Rank(A^T)

Q68. [GATE CE 2017] NAT | Moderate

Rank of the matrix [2, 3, 4; 4, 3, 1; 1, 2, 4] is _____.

Q69. [GATE DA Expected 2027] NAT | Difficult

Let A be a 6×4 matrix and B be a 4×6 matrix such that Rank(A) = 4 and Rank(B) = 4. What is the nullity of the 6×6 matrix AB?

Q70. [Application-Based] MCQ | Moderate

In low-rank matrix factorization (e.g. recommendation systems), a rating matrix $R \in \mathbb{R}^{m \times n}$ is approximated by $U V^T$ where $U \in \mathbb{R}^{m \times r}$ and $V \in \mathbb{R}^{n \times r}$. The maximum rank of the reconstructed matrix $U V^T$ is:

(A) r
(B) min(m, n)
(C) $m + n$
(D) r^2

Q71. [GATE CS 2011] NAT | Easy

The rank of the matrix
[1, 1, 1; 1, −1, 0; 1, 1, 1] is _____.

Q72. [Advanced Conceptual] NAT | Moderate

Let A be a real 5×3 matrix and B be a real 3×5 matrix such that Rank(AB) = 3. What is the value of Rank(BA)?

Q73. [GATE IN 2017] NAT | Moderate

If Rank of [1, 2, 3; 4, 5, 6; 7, 8, k] is 2, then the value of k is _____.

Q74. [Mixed Concept] MCQ | Difficult

Let A be an $n \times n$ matrix. Which of the following is equivalent to stating that Rank(A) = n?

(A) The column vectors of A form a basis for $\mathbb{R}^n$
(B) The homogeneous equation $Ax = 0$ has non-trivial solutions
(C) det(A) = 0
(D) 0 is an eigenvalue of A

Q75. [GATE DA Expected 2027] NAT | Moderate

Let A and B be 4×4 matrices such that Rank(A) = 3 and Rank(B) = 2. The maximum possible rank of $A + B$ is _____.

Q76. [GATE EC 2015] NAT | Easy

The rank of the matrix [0, 0, 0; 0, 0, 0; 0, 0, 0] is _____.

Q77. [Advanced Conceptual]

MSQ | Difficult

Let $A \in \mathbb{R}^{m \times n}$. Which of the following vector spaces have dimension equal to the rank of A?

(A) Row space of A
(B) Column space of A
(C) Null space of A
(D) Left null space of A

Q78. [GATE DA Expected 2027]

NAT | Difficult

Let A be a 7×5 matrix with rank 4, and B be a 5×8 matrix with rank 3. The minimum possible rank of AB is _____.

Q79. [Application-Based]

NAT | Moderate

In collaborative filtering, a matrix of 100 users and 50 movies is formed by the sum of two rank-1 matrices. The rank of this movie rating matrix is _____.

Q80. [GATE CS 2007]

NAT | Moderate

The rank of the matrix [1, 2, 3; 4, 5, 6; 7, 8, 9] is _____.

CHAPTER 4: SYSTEMS OF LINEAR EQUATIONS (Q81 – Q110)

Q81. [GATE DA 2024, Question 41]

MCQ | Moderate

Consider the linear system $Ax = b$, where $A \in \mathbb{R}^{m \times n}$. The system has at least one solution for EVERY $b \in \mathbb{R}^m$ if and only if:

(A) Rank(A) = m
(B) Rank(A) = n
(C) $m = n$
(D) Rank(A) < m

Q82. [GATE CS 2021, Set 2]

NAT | Moderate

For what value of c does the system of equations:

$x + y + z = 6$

$x + 2y + 3z = 10$

$x + 2y + c\,z = 10$

have infinitely many solutions?

Q83. [GATE EC 2019]

NAT | Moderate

Consider the system:

$2x + y + z = 4$

$x + 2y + z = 4$

$x + y + 2z = 4$

The value of $x + y + z$ is _____.

Q84. [Advanced Conceptual]

MSQ | Difficult

Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Which of the following conditions is/are NECESSARY AND SUFFICIENT for the system $Ax = b$ to have a UNIQUE solution?

(A) Rank(A) = Rank([$A \mid b$]) = n
(B) Nullity(A) = 0 and b is in the column space of A
(C) $m = n$ and det(A) ≠ 0
(D) $m \geq n$ and Rank(A) = n (for all b)

Q85. [GATE DA Expected 2027]

NAT | Difficult

Consider the linear system of equations:

$x + y + z = 1$

$ax + by + cz = d$

$a^2x + b^2y + c^2z = d^2$

with $a = 1$, $b = 2$, $c = 3$, and $d = 4$. The value of z is _____.

Q86. [GATE ME 2020]

MCQ | Easy

A system of m linear equations in n unknowns has a unique solution if:

(A) Rank(A) = Rank([A | b]) = n
(B) Rank(A) = Rank([A | b]) < n
(C) Rank(A) < Rank([A | b])
(D) $m < n$

Q87. [GATE CS 2017]

NAT | Moderate

The number of linearly independent solutions to the homogeneous linear system

$x_1 + 2x_2 + 3x_3 + 4x_4 = 0$

$2x_1 + 4x_2 + 6x_3 + 8x_4 = 0$

is _____.

Q88. [Application-Based]

MCQ | Moderate

In ordinary least-squares (OLS) linear regression, the normal equations to solve for weight vector w given feature matrix X and target y are:

(A) $X^TX\,w = X^Ty$
(B) $X\,X^T\,w = X\,y$
(C) $X\,w = y$
(D) $X^T\,w = X\,y$

Q89. [GATE EE 2016]

NAT | Easy

For the system $x + 2y = 3$ and $2x + 4y = k$, for what value of k is the system consistent?

Q90. [Advanced Conceptual]

NAT | Difficult

Let A be a 4×3 matrix and b be a vector such that $Ax = b$ has NO solution. What is the maximum possible rank of matrix A?

Q91. [GATE CE 2018]

MCQ | Moderate

If Rank(A) < Rank([A | b]), then the system $Ax = b$ has:

(A) No solution
(B) Unique solution
(C) Infinitely many solutions
(D) Trivial solution

Q92. [GATE DA Expected 2027]

NAT | Moderate

Consider the homogeneous system $Ax = 0$ where A is a 5×7 matrix with Rank(A) = 4. The dimension of the solution space is _____.

Q93. [Mixed Concept]

MSQ | Difficult

Let $Ax = b$ be a linear system with $A \in \mathbb{R}^{m \times n}$. Let x_p be a particular solution and x_h be any solution to the homogeneous equation $Ax = 0$. Which of the following is/are TRUE?

(A) $x_p + x_h$ is always a solution to $Ax = b$
(B) The difference of any two solutions of $Ax = b$ lies in the null space of A
(C) The set of all solutions of $Ax = b$ forms a vector subspace of $\mathbb{R}^n$ for all $b \neq 0$
(D) If $m < n$, the system can never have a unique solution

Q94. [GATE IN 2019]

NAT | Moderate

For the system

$x + y + z = 3$

$x + 2y + 3z = 6$

$x + 4y + 9z = 14$

the value of y is _____.

Q95. [GATE CS 2013]

NAT | Easy

If $Ax = 0$ has only the trivial solution $x = 0$ for an $n \times n$ matrix A, then Rank(A) is equal to _____ (in terms of n, for $n = 5$).

Q96. [GATE DA Expected 2027]

NAT | Difficult

The value of λ for which the system:

$2x - y + 2z = 2$

$x - 2y + z = -4$

$x + y + \lambda z = 4$

has NO solution is _____.

Q97. [Application-Based]

NAT | Moderate

In polynomial curve fitting, fitting a parabola $y = a\,x^2 + b\,x + c$ through points (1, 1), (2, 4), and (3, 9) yields the coefficient a equal to _____.

Q98. [GATE EC 2014]

MCQ | Easy

For what condition does a homogeneous system of n linear equations in n variables have a non-trivial solution?

(A) det(A) = 0
(B) det(A) ≠ 0
(C) Rank(A) = n
(D) Trace(A) = 0

Q99. [Advanced Conceptual]

MSQ | Difficult

Let $A \in \mathbb{R}^{m \times n}$ with $m > n$. Which of the following is/are TRUE?

(A) If Rank(A) = n, then $Ax = b$ has either 0 or 1 solution for any b
(B) A^TA is invertible if and only if Rank(A) = n
(C) The system $Ax = b$ always has a solution for every $b \in \mathbb{R}^m$
(D) The least-squares solution to $Ax = b$ minimizes $||Ax - b||_2$

Q100. [GATE ME 2014]

NAT | Moderate

For the system $x + 2y + z = 4$, $2x + y + 2z = 5$, $x - y + z = 1$, the rank of the augmented matrix is _____.

Q101. [GATE DA Expected 2027] NAT | Moderate

Consider the underdetermined system with 4 variables and 2 equations:

$x_1 + x_2 + x_3 + x_4 = 4$

$x_1 + 2x_2 + 3x_3 + 4x_4 = 10$

The number of free variables in the solution parametrization is _____.

Q102. [GATE CS 2004] MCQ | Moderate

The system of equations $x + y + z = 6$, $x + 4y + 6z = 20$, $x + 4y + \lambda z = \mu$ has infinitely many solutions when:

(A) $\lambda = 6, \mu = 20$
(B) $\lambda = 6, \mu \neq 20$
(C) $\lambda \neq 6, \mu = 20$
(D) $\lambda \neq 6, \mu \neq 20$

Q103. [Mixed Concept] NAT | Difficult

Let $A = [1, 2; 2, 4]$ and $b = [3, 6]^T$. The minimum Euclidean norm $||x||_2$ of a solution vector $x = [x_1, x_2]^T$ satisfying $Ax = b$ is $3 / \sqrt{k}$. The value of integer k is _____.

Q104. [GATE CE 2015] NAT | Easy

The solution to the system $2x + 3y = 7$, $3x + 2y = 8$ gives $x =$ _____.

Q105. [Advanced Conceptual] NAT | Moderate

Let A be a 3×3 matrix such that A [1; 1; 1] = [1; 1; 1] and A [1; 2; 3] = [0; 0; 0]. If Rank(A) = 2, then the nullity of A is _____.

Q106. [Application-Based] MCQ | Moderate

In Ridge Regression (Tikhonov Regularization), the linear system solved is $(X^TX + \lambda I)\, w = X^Ty$ with $\lambda > 0$. For any real matrix X, the matrix $X^TX + \lambda I$ is:

(A) Always strictly positive definite and invertible
(B) Singular if X has dependent columns
(C) Skew-symmetric
(D) Orthogonal

Q107. [GATE EE 2014] NAT | Easy

If A is a 3×3 non-singular matrix, the number of solutions to $Ax = 0$ is _____.

Q108. [GATE DA Expected 2027] MSQ | Difficult

Consider the system $Ax = b$. Which of the following is/are equivalent to the Fredholm Alternative?

(A) $Ax = b$ has a solution if and only if $y^Tb = 0$ for all y such that $A^Ty = 0$
(B) Column space of A is the orthogonal complement of the null space of A^T
(C) Either $Ax = b$ has a solution or there exists y such that $A^Ty = 0$ and $y^Tb \neq 0$
(D) $Ax = b$ always has unique solution for any square matrix A

Q109. [GATE CS 2003] NAT | Moderate

For the system $x + y = 2$, $2x + 2y = 4$, the number of solutions is _____ (enter 999 if infinitely many).

Q110. [Mixed Concept] NAT | Difficult

Let A be a 3×3 matrix with eigenvalues 1, 2, 3. The number of distinct solution vectors to $Ax = [1, 2, 3]^T$ is _____.

CHAPTER 5: VECTOR SPACES & SUBSPACES (Q111 – Q135)

Q111. [GATE DA 2024, Question 18] `MSQ | Moderate`

Which of the following subsets of $\mathbb{R}^3$ is/are valid vector subspaces of $\mathbb{R}^3$ over $\mathbb{R}$?

(A) $W_1 = \{[x, y, z]^T \in \mathbb{R}^3 : 2x - 3y + z = 0\}$
(B) $W_2 = \{[x, y, z]^T \in \mathbb{R}^3 : x + y + z = 1\}$
(C) $W_3 = \{[x, y, z]^T \in \mathbb{R}^3 : xy = 0\}$
(D) $W_4 = \{[x, y, z]^T \in \mathbb{R}^3 : x = 2y \text{ and } z = 3y\}$

Q112. [GATE CS 2016] `MCQ | Moderate`

Let V be the set of all 2×2 real symmetric matrices. Then V is a vector space over $\mathbb{R}$ of dimension:

(A) 3
(B) 4
(C) 2
(D) 1

Q113. [Advanced Conceptual] `MSQ | Difficult`

Let W_1 and W_2 be subspaces of a vector space V. Which of the following is/are ALWAYS a subspace of V?

(A) $W_1 \cap W_2$
(B) $W_1 + W_2 = \{w_1 + w_2 : w_1 \in W_1, w_2 \in W_2\}$
(C) $W_1 \cup W_2$
(D) $W_1 \setminus W_2$

Q114. [GATE DA Expected 2027] `NAT | Moderate`

Let V be the vector space of all polynomials in x with real coefficients of degree at most 4. The dimension of V is _____.

Q115. [GATE MA 2021] `NAT | Difficult`

Let $W = \{A \in \mathbb{R}^{3\times3} : A^T = -A \text{ and } \text{trace}(A) = 0\}$. The dimension of W over $\mathbb{R}$ is _____.

Q116. [Application-Based] `MCQ | Moderate`

In signal processing, the set of all band-limited continuous functions satisfying $f(t) = f(-t)$ forms:

(A) A subspace of the space of continuous functions
(B) Not a subspace because zero function is absent
(C) A finite set of functions
(D) A non-convex cone

Q117. [GATE CS 2014] `MCQ | Easy`

Which of the following sets with standard operations is NOT a vector space over $\mathbb{R}$?

(A) Set of all 2×2 matrices with determinant = 0
(B) Set of all continuous real-valued functions on [0, 1]
(C) Set of all convergent sequences of real numbers
(D) Set of all solutions to $y'' + 3y' + 2y = 0$

Q118. [Advanced Conceptual] `NAT | Difficult`

Let $V = \mathbb{R}^5$. Let W_1 be a subspace of dimension 3 and W_2 be a subspace of dimension 4. What is the MINIMUM possible dimension of $W_1 \cap W_2$?

Q119. [GATE DA Expected 2027]

MSQ | Difficult

Let V be the vector space of all $n \times n$ real matrices. Which of the following subsets is/are subspaces of V?

(A) Set of all matrices with trace equal to 0
(B) Set of all diagonal matrices
(C) Set of all upper triangular matrices
(D) Set of all invertible matrices

Q120. [GATE EC 2020]

NAT | Easy

Dimension of the vector space of all 3×3 diagonal matrices over $\mathbb{R}$ is _____.

Q121. [Mixed Concept]

NAT | Moderate

Let $W_1 = \{[x, y, z]^T : x + y + z = 0\}$ and $W_2 = \{[x, y, z]^T : x - y + 2z = 0\}$ in $\mathbb{R}^3$. The dimension of $W_1 \cap W_2$ is _____.

Q122. [GATE MA 2019]

NAT | Moderate

The dimension of the vector space of all 4×4 real skew-symmetric matrices is _____.

Q123. [Advanced Conceptual]

MCQ | Moderate

Under what condition is the union of two subspaces $W_1 \cup W_2$ a subspace?

(A) Either $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$
(B) Always
(C) If $W_1 \cap W_2 = \{0\}$
(D) Never

Q124. [GATE DA Expected 2027]

NAT | Difficult

Let V be the vector space of all real polynomials of degree ≤ 3. Let $W = \{p(x) \in V : p(1) = 0 \text{ and } p'(1) = 0\}$. The dimension of W is _____.

Q125. [Application-Based]

MSQ | Moderate

In representation learning and autoencoders, the latent space is a linear subspace of $\mathbb{R}^D$. If latent dimension is $d < D$, which properties hold?

(A) It contains the origin (0 vector)
(B) It is closed under linear combinations
(C) Its orthogonal complement has dimension $D - d$
(D) It can capture non-linear manifold topologies without kernels

Q126. [GATE CS 2009]

NAT | Easy

Dimension of the vector space $\mathbb{R}^4$ over $\mathbb{R}$ is _____.

Q127. [Advanced Conceptual]

NAT | Moderate

Let V be the vector space of $n \times n$ matrices. The dimension of the subspace of trace-zero matrices is $n^2 - c$. The value of c is _____.

Q128. [GATE DA Expected 2027]

MCQ | Moderate

Let $S = \{[x, y]^T \in \mathbb{R}^2 : x \geq 0, y \geq 0\}$ (the first quadrant). Which vector space axiom FAILS for S?

(A) Existence of additive inverse
(B) Additive associativity
(C) Additive commutativity
(D) Closure under positive scalar multiplication

Q129. [Mixed Concept] NAT | Difficult

Let W_1 and W_2 be two distinct 4-dimensional subspaces of $\mathbb{R}^6$. What is the maximum possible dimension of $W_1 \cap W_2$?

Q130. [GATE MA 2018] NAT | Moderate

Let V be the vector space of all 3×3 matrices. Dimension of the subspace of all symmetric matrices with trace 0 is _____.

Q131. [GATE CS 2006] NAT | Easy

The zero vector space $\{0\}$ has dimension _____.

Q132. [Advanced Conceptual] MSQ | Difficult

Let V be an inner product space. For any subset $S \subseteq V$, which of the following is/are TRUE about the orthogonal complement $S^\perp$?

(A) $S^\perp$ is always a subspace of V, even if S is not
(B) $S \cap S^\perp \subseteq \{0\}$
(C) $(S^\perp)^\perp = span(S)$ (in finite dimensions)
(D) $S^\perp = \{0\}$ implies $S = V$

Q133. [GATE DA Expected 2027] NAT | Moderate

Let P_3 denote polynomials of degree at most 3. The dimension of the subspace $W = \{p \in P_3 : p(2) = 0\}$ is _____.

Q134. [Application-Based] MCQ | Moderate

In support vector machines (SVM), a hyperplane separating two classes in $\mathbb{R}^d$ is an affine subspace of dimension:

(A) $d - 1$
(B) d
(C) 1
(D) $d / 2$

Q135. [Mixed Concept] NAT | Difficult

Let A be a 3×4 matrix. The dimension of the row space plus the dimension of the null space of A is _____.

CHAPTER 6: LINEAR INDEPENDENCE & DEPENDENCE (Q136 – Q160)

Q136. [GATE DA 2024, Question 12] NAT | Moderate

For what value of α are the vectors $v_1 = [1, 2, 3]^T$, $v_2 = [0, 1, 2]^T$, and $v_3 = [2, 3, \alpha]^T$ linearly dependent?

Q137. [GATE CS 2018] MCQ | Easy

Which of the following sets of vectors in $\mathbb{R}^3$ is linearly independent?

(A) $\{[1, 0, 0]^T, [0, 1, 0]^T, [1, 1, 0]^T\}$
(B) $\{[1, 2, 3]^T, [2, 4, 6]^T\}$
(C) $\{[1, 0, 0]^T, [0, 1, 0]^T, [0, 0, 1]^T\}$
(D) $\{[0, 0, 0]^T, [1, 2, 3]^T\}$

Q138. [GATE EC 2016] NAT | Moderate

The number of linearly independent vectors in the set $\{[1, 1, 0]^T, [0, 1, 1]^T, [1, 0, 1]^T, [1, 1, 1]^T\}$ in $\mathbb{R}^3$ is _____.

Q139. [Advanced Conceptual] MSQ | Difficult

Let $\{v_1, v_2, v_3\}$ be a linearly independent set of vectors in a vector space V. Which of the following sets is/are ALSO GUARANTEED to be linearly independent?

(A) $\{v_1 + v_2, v_2 + v_3, v_3 + v_1\}$
(B) $\{v_1 - v_2, v_2 - v_3, v_3 - v_1\}$
(C) $\{v_1, v_1 + v_2, v_1 + v_2 + v_3\}$
(D) $\{2v_1, 3v_2, 4v_3\}$

Q140. [GATE DA Expected 2027] NAT | Difficult

Consider the functions $f_1(x) = 1, f_2(x) = \sin^2(x), f_3(x) = \cos^2(x)$. The rank of the Wronskian-type matrix formed by these functions on $\mathbb{R}$ is _____.

Q141. [Application-Based] MCQ | Moderate

In deep neural networks, if the columns of a weight matrix $W \in \mathbb{R}^{d \times m}$ are linearly dependent, this phenomenon is called:

(A) Feature redundancy (rank deficiency)
(B) Exploding gradients
(C) Vanishing activation
(D) Over-regularization

Q142. [GATE ME 2017] NAT | Easy

Any set containing the zero vector 0 is always linearly _____ (enter 1 for independent, 0 for dependent).

Q143. [GATE MA 2020] NAT | Moderate

Consider polynomials $1 + x, x + x^2$, and $1 + x^2$ in $P_2(\mathbb{R})$. The number of linearly independent polynomials among them is _____.

Q144. [Advanced Conceptual] MCQ | Difficult

If $S = \{v_1, \ldots, v_k\}$ is a linearly independent set in $\mathbb{R}^n$, then:

(A) $k \leq n$ must always hold
(B) $k \geq n$
(C) $k = n$
(D) No restriction exists on k

Q145. [GATE DA Expected 2027] NAT | Moderate

Let $v_1 = [1, 1, 1]^T, v_2 = [1, 2, 3]^T, v_3 = [1, 4, k]^T$. For what value of k are these vectors linearly dependent?

Q146. [GATE CS 2013] NAT | Easy

Maximum number of linearly independent vectors in $\mathbb{R}^3$ is _____.

Q147. [Mixed Concept]

MSQ | Difficult

Let A be an $m \times n$ matrix. Which of the following statements is/are equivalent to: "The columns of A are linearly independent"?

(A) $Ax = 0$ has only the trivial solution $x = 0$
(B) Rank(A) = n
(C) A^TA is an invertible $n \times n$ matrix
(D) Nullity(A) = 0

Q148. [GATE CE 2019]

NAT | Moderate

Let $v_1 = [2, 1]^T$ and $v_2 = [k, 3]^T$. If v_1 and v_2 are collinear, then $k =$ _____.

Q149. [GATE DA Expected 2027]

NAT | Difficult

Consider the set of 2×2 matrices: $M_1 = [1, 0; 0, 0]$, $M_2 = [0, 1; 1, 0]$, $M_3 = [0, 0; 0, 1]$, $M_4 = [1, 2; 2, 1]$. How many linearly independent matrices are in this set?

Q150. [Application-Based]

MCQ | Moderate

In multi-collinearity detection in regression, the Variance Inflation Factor (VIF) becomes infinite when:

(A) One predictor feature is an exact linear combination of others
(B) All predictors are completely orthogonal
(C) Sample size $N >> d$
(D) Target variable is binary

Q151. [GATE EE 2018]

NAT | Easy

If two vectors $u = [1, a]^T$ and $v = [3, 6]^T$ are linearly dependent, then $a =$ _____.

Q152. [Advanced Conceptual]

MSQ | Difficult

Let S be a subset of a vector space V. Which of the following is/are TRUE?

(A) Any non-empty subset of a linearly independent set is linearly independent
(B) Any superset of a linearly dependent set is linearly dependent
(C) A single non-zero vector is always linearly independent
(D) If $span(S) = V$, then S must be linearly independent

Q153. [GATE DA Expected 2027]

NAT | Moderate

Let $f(x) = e^x$, $g(x) = e^{2x}$, $h(x) = e^{3x}$. At $x = 0$, the determinant of the Wronskian matrix $W(f, g, h)(0)$ is _____.

Q154. [GATE CS 2004]

NAT | Easy

Number of linearly independent eigenvectors of a 3×3 identity matrix I_3 is _____.

Q155. [Mixed Concept]

NAT | Difficult

Let $v_1, v_2, v_3 \in \mathbb{R}^3$ be mutually orthogonal non-zero vectors. The dimension of their span is _____.

Q156. [GATE IN 2016]

NAT | Moderate

For the vectors $[1, 2, 1]^T$, $[2, 1, 3]^T$, $[7, 5, k]^T$ to be linearly dependent, the value of k is _____.

Q157. [Advanced Conceptual] MCQ | Moderate

Let $\{u, v\}$ be linearly independent. Then $\{u + v, u - v\}$ is linearly dependent:

(A) Over fields of characteristic 2
(B) Over $\mathbb{R}$
(C) Over $\mathbb{C}$
(D) Never

Q158. [GATE DA Expected 2027] NAT | Moderate

How many vectors in the set $\{[1, 0, 0]^T, [0, 1, 0]^T, [0, 0, 1]^T, [1, 2, 3]^T\}$ can be chosen at most to form a linearly independent set?

Q159. [Application-Based] NAT | Moderate

In a dataset with 5 features, if $X_5 = 2X_1 - 3X_2$ and $X_4 = X_2 + X_3$, while X_1, X_2, X_3 are independent, the number of linearly independent feature columns is _____.

Q160. [GATE ME 2013] NAT | Easy

The number of linearly independent rows in a 3×3 singular matrix cannot exceed _____.

CHAPTER 7: BASIS & DIMENSION (Q161 – Q185)

Q161. [GATE DA 2024, Question 28] NAT | Moderate

Let V be the subspace of $\mathbb{R}^4$ spanned by $[1, 1, 0, 1]^T$, $[0, 1, 1, 0]^T$, $[1, 2, 1, 1]^T$, and $[1, 0, -1, 1]^T$. The dimension of V is _____.

Q162. [GATE CS 2017] NAT | Easy

The dimension of the null space of matrix
$A = [1, 2, 3; 4, 5, 6; 7, 8, 9]$ is _____.

Q163. [GATE MA 2021] NAT | Difficult

Let $V = \{p(x) \in \mathbb{R}[x] : deg(p) \leq 3, p(0) = 0, p'(1) = 0\}$. The dimension of V as a vector space over $\mathbb{R}$ is _____.

Q164. [Advanced Conceptual] MSQ | Difficult

Let V be an n-dimensional vector space. Which of the following is/are TRUE?

(A) Any linearly independent set of n vectors in V is a basis
(B) Any spanning set of n vectors in V is a basis
(C) Every basis of V has exactly n vectors
(D) Any set of $n + 1$ vectors in V must be linearly dependent

Q165. [GATE DA Expected 2027] NAT | Difficult

Let W_1 and W_2 be subspaces of $\mathbb{R}^8$ with $\dim(W_1) = 5$ and $\dim(W_2) = 5$. If $\dim(W_1 + W_2) = 7$, then the dimension of $W_1 \cap W_2$ is _____.

Q166. [Application-Based]

MCQ | Moderate

In Fourier analysis and discrete cosine transforms (DCT) used in data compression, the basis functions are chosen because they form:

(A) An orthonormal basis for the signal space
(B) A linearly dependent spanning set
(C) A non-orthogonal basis with arbitrary angles
(D) A set with zero inner products everywhere

Q167. [GATE EC 2018]

NAT | Easy

If A is a 4×3 matrix with rank 2, the dimension of the null space of A is _____.

Q168. [GATE CS 2015]

NAT | Moderate

The dimension of the solution space of $x_1 + x_2 + x_3 + x_4 = 0$ in $\mathbb{R}^4$ is _____.

Q169. [Advanced Conceptual]

NAT | Difficult

Let $M_{3\times3}(\mathbb{R})$ be the vector space of all 3×3 matrices. Let S be the subspace of symmetric matrices and K be the subspace of skew-symmetric matrices. The dimension of $S \cap K$ is _____.

Q170. [GATE DA Expected 2027]

MSQ | Difficult

Let $B = \{v_1, v_2, v_3\}$ be an ordered basis for $\mathbb{R}^3$. If a linear transformation $T: \mathbb{R}^3 \to \mathbb{R}^3$ satisfies $T(v_1) = v_2$, $T(v_2) = v_3$, $T(v_3) = 0$, which of the following is/are TRUE?

(A) $T^3 = 0$ (nilpotent of index 3)
(B) Rank(T) = 2
(C) Nullity(T) = 1
(D) All eigenvalues of T are 0

Q171. [GATE EE 2019]

NAT | Easy

The dimension of the row space of a 5×3 matrix of rank 3 is _____.

Q172. [Mixed Concept]

NAT | Difficult

Let A = [1, 2, 1; 2, 4, 2; 3, 6, 3]. The dimension of the orthogonal complement of the column space of A in $\mathbb{R}^3$ is _____.

Q173. [GATE ME 2018]

MCQ | Moderate

A basis for a vector space V is defined as:

(A) A linearly independent spanning set of V
(B) Any set that spans V
(C) Any linearly independent set in V
(D) An orthogonal set of unit vectors

Q174. [GATE DA Expected 2027]

NAT | Moderate

Let V be the vector space of all 3×3 upper triangular matrices. The dimension of V is _____.

Q175. [Application-Based]

NAT | Moderate

In dictionary learning for sparse coding, an overcomplete dictionary has $K = 100$ basis atoms in $\mathbb{R}^{20}$. The maximum dimension of the subspace spanned by this dictionary is _____.

Q176. [GATE IN 2020] NAT | Easy

If a matrix $A \in \mathbb{R}^{3\times5}$ has rank 2, the dimension of the null space of A^T is _____.

Q177. [Advanced Conceptual] MSQ | Difficult

Let V and W be finite-dimensional vector spaces and $T: V \to W$ be a linear transformation. Which of the following is/are TRUE?

(A) dim(ker T) + dim(range T) = dim V
(B) If dim V = dim W, T is injective if and only if T is surjective
(C) dim(range T) ≤ min(dim V, dim W)
(D) ker T is a subspace of W

Q178. [GATE CE 2016] NAT | Moderate

Dimension of the vector space of all 2×2 matrices with real entries is _____.

Q179. [GATE DA Expected 2027] NAT | Difficult

Let $W = \{[x, y, z, w]^T \in \mathbb{R}^4 : x + 2y - z = 0,\ 2x + y + w = 0\}$. The dimension of W is _____.

Q180. [GATE CS 2008] NAT | Easy

The standard basis of $\mathbb{R}^2$ consists of _____ vectors.

Q181. [Mixed Concept] NAT | Difficult

Let P_n denote polynomials of degree ≤ n. The dimension of the quotient space P_4 / P_2 is _____.

Q182. [GATE EC 2013] MCQ | Moderate

Which of the following forms a basis for $\mathbb{R}^2$?

(A) $\{[1, 2]^T, [2, 4]^T\}$
(B) $\{[1, 0]^T, [1, 1]^T\}$
(C) $\{[1, 1]^T, [2, 2]^T, [3, 3]^T\}$
(D) $\{[0, 0]^T, [1, 2]^T\}$

Q183. [Advanced Conceptual] NAT | Moderate

If B_1 and B_2 are two distinct bases for a vector space V with $|B_1| = 7$, then $|B_2|$ is _____.

Q184. [Application-Based] NAT | Moderate

In word embedding models (e.g. Word2Vec) mapping words to 300-dimensional vectors, the maximum number of mutually orthogonal semantic concept axes is _____.

Q185. [GATE ME 2011] NAT | Easy

The dimension of the vector space of all scalar multiples of a single non-zero vector in $\mathbb{R}^3$ is _____.

CHAPTER 8: EIGENVALUES & EIGENVECTORS (Q186 – Q215)

Q186. [GATE DA 2024, Question 8] NAT | Moderate

Let $A = [4, 2; 1, 3]$. The sum of the eigenvalues of $A^2 + 2A$ is _____.

Q187. [GATE CS 2022] NAT | Moderate

Let A be a 3×3 real matrix with eigenvalues 1, −1, 0. The determinant of $I + A^{100}$ is _____.

Q188. [GATE EC 2022] NAT | Easy

For the matrix $A = [2, 1; 1, 2]$, the largest eigenvalue is _____.

Q189. [Advanced Conceptual] MSQ | Difficult

Let A be an $n \times n$ real symmetric matrix. Which of the following statements is/are ALWAYS TRUE?

(A) All eigenvalues of A are purely real
(B) Eigenvectors corresponding to distinct eigenvalues are mutually orthogonal
(C) The geometric multiplicity of each eigenvalue equals its algebraic multiplicity
(D) A can be diagonalized by an orthogonal matrix

Q190. [GATE DA Expected 2027] NAT | Difficult

Let A be a 3×3 matrix with characteristic polynomial $p(\lambda) = \lambda^3 - 6\lambda^2 + 11\lambda - 6$. The trace of A^{-1} is p / q in lowest terms. The value of $p + q$ is _____.

Q191. [Application-Based] MCQ | Moderate

In Google's PageRank algorithm, the transition probability matrix P is stochastic. By Perron-Frobenius theorem, the dominant eigenvalue of P is:

(A) 1
(B) 0
(C) Dependent on graph size
(D) ∞

Q192. [GATE CS 2020] NAT | Moderate

Let $A = [1, 2; 0, 2]$. One of the eigenvectors of A corresponding to $\lambda = 2$ is $[2, k]^T$. The value of k is _____.

Q193. [GATE EE 2020] NAT | Easy

The product of the eigenvalues of the matrix
$[1, 0, 2; 0, 3, 0; 2, 0, 1]$ is _____.

Q194. [Advanced Conceptual] NAT | Difficult

Let A be a 3×3 matrix with eigenvalues 2, 3, 5. The determinant of matrix $B = A^2 - 5A + 6I$ is _____.

Q195. [GATE DA Expected 2027] MSQ | Difficult

Let A be a 4×4 skew-symmetric real matrix. Which of the following is/are TRUE?

(A) All non-zero eigenvalues are purely imaginary
(B) If λ is an eigenvalue, then $-\lambda$ is also an eigenvalue
(C) $\det(A) \geq 0$
(D) $\text{Trace}(A) = 0$

Q196. [GATE ME 2019] NAT | Easy

The sum of the eigenvalues of $[3, 4; 4, -3]$ is _____.

Q197. [GATE CE 2021] NAT | Moderate

If λ_1, λ_2 are eigenvalues of $[5, 4; 1, 2]$, the value of $|\lambda_1 - \lambda_2|$ is _____.

Q198. [Mixed Concept] NAT | Difficult

Let $A = [0, 1, 0; 0, 0, 1; 1, 0, 0]$. The magnitude of each of the three complex eigenvalues of A is _____.

Q199. [GATE CS 2017] NAT | Moderate

For the matrix $A = [1, -1; 2, 4]$, if $v = [1, k]^T$ is an eigenvector corresponding to eigenvalue 2, then $k =$ _____.

Q200. [GATE DA Expected 2027] NAT | Difficult

Let A be a 3×3 matrix whose row sums are all equal to 7. What is one guaranteed eigenvalue of A?

Q201. [Application-Based] MCQ | Moderate

In Spectral Graph Theory, the graph Laplacian matrix $L = D - A$ of a connected graph always has its smallest eigenvalue equal to:

(A) 0
(B) 1
(C) −1
(D) Vertex degree

Q202. [GATE IN 2019] NAT | Easy

The eigenvalues of a diagonal matrix are simply its _____ entries (enter 1 if true, 0 if false).

Q203. [Advanced Conceptual] MSQ | Difficult

Let $A \in \mathbb{R}^{n \times n}$ with eigenvalue λ and eigenvector x. Which of the following is/are TRUE?

(A) For any scalar c, $A + cI$ has eigenvalue $\lambda + c$ with eigenvector x
(B) A^k has eigenvalue λ^k with eigenvector x
(C) If A is invertible, A^{-1} has eigenvalue $1/\lambda$ with eigenvector x
(D) A^T has the same eigenvalues as A with the exact same eigenvectors x

Q204. [GATE ME 2017] NAT | Moderate

The minimum eigenvalue of the matrix [3, 5; 5, 3] is _____.

Q205. [GATE DA Expected 2027] NAT | Difficult

Let $u = [1, 1, 1, 1]^T$. Let $A = u\,u^T$. The non-zero eigenvalue of A is _____.

Q206. [GATE EC 2017] NAT | Easy

Eigenvalues of [0, 1; −1, 0] are $\pm j\,k$. The positive real number $k =$ _____.

Q207. [Mixed Concept] NAT | Difficult

Let A be a 3×3 matrix with eigenvalues 1, 2, 4. The trace of adj(A) is _____.

Q208. [GATE CS 2011] NAT | Moderate

If the characteristic equation of a 3×3 matrix is $\lambda^3 - \lambda^2 - \lambda - 1 = 0$, then det($A$) is _____.

Q209. [Advanced Conceptual] MCQ | Moderate

By the Cayley-Hamilton Theorem, every square matrix satisfies its own:

(A) Characteristic equation
(B) Adjoint relation
(C) Trace decomposition
(D) Norm equation

Q210. [GATE DA Expected 2027] NAT | Moderate

For the matrix $A = [1, 2; 3, 2]$, using Cayley-Hamilton theorem, $A^2 - 3A = k\,I$. The value of k is _____.

Q211. [Application-Based] NAT | Moderate

In stability analysis of continuous dynamical systems $dx/dt = Ax$, the system is asymptotically stable if all eigenvalues of A have real parts strictly less than _____.

Q212. [GATE EE 2013] NAT | Easy

The number of distinct eigenvalues of the matrix [2, 0; 0, 2] is _____.

Q213. [Advanced Conceptual] MSQ | Difficult

Let A be a square matrix. If λ is an eigenvalue of A with algebraic multiplicity $AM(\lambda)$ and geometric multiplicity $GM(\lambda)$, then:

(A) $1 \leq GM(\lambda) \leq AM(\lambda)$ always
(B) $GM(\lambda) = n - Rank(A - \lambda I)$
(C) If $GM(\lambda) < AM(\lambda)$, A is defective (non-diagonalizable)
(D) $GM(\lambda)$ can strictly exceed $AM(\lambda)$

Q214. [GATE CE 2012] NAT | Easy

For a 2×2 matrix with eigenvalues -2 and -3, the trace is _____.

Q215. [Mixed Concept] NAT | Difficult

Let A = [2, 1, 1; 1, 2, 1; 1, 1, 2]. The largest eigenvalue of A is _____.

CHAPTER 9: DIAGONALIZATION (Q216 – Q235)

Q216. [GATE DA 2024, Question 31] MCQ | Moderate

Which of the following matrices is NOT diagonalizable over $\mathbb{R}$?

(A) [1, 1; 0, 1]
(B) [1, 1; 1, 1]
(C) [2, 0; 0, 3]
(D) [1, 2; 2, 1]

Q217. [GATE CS 2019] MCQ | Moderate

A matrix $A \in \mathbb{R}^{n \times n}$ is diagonalizable if and only if:

(A) It has n linearly independent eigenvectors
(B) It has n distinct eigenvalues
(C) $\det(A) \neq 0$
(D) A is symmetric

Q218. [GATE EC 2015] NAT | Moderate

Let A = [5, 3; 3, 5]. If $P^{-1}AP = D = [d_1, 0; 0, d_2]$ with $d_1 > d_2$, then d_1 = _____.

Q219. [Advanced Conceptual] MSQ | Difficult

Which of the following classes of matrices are ALWAYS guaranteed to be diagonalizable over $\mathbb{C}$?

(A) Real symmetric matrices
(B) Matrices with all distinct eigenvalues
(C) Unitary and Orthogonal matrices
(D) Non-zero Nilpotent matrices

Q220. [GATE DA Expected 2027] NAT | Difficult

Let A = [2, 0, 0; 0, 3, 1; 0, 0, 3]. The geometric multiplicity of eigenvalue 3 is _____.

Q221. [Application-Based] MCQ | Moderate

In discrete dynamical state updates $x_{k+1} = A\,x_k$, computing A^k is drastically simplified when $A = P\,D\,P^{-1}$ because:

(A) $A^k = P\,D^k\,P^{-1}$ where D^k simply powers diagonal entries
(B) $A^k = D^k$ always
(C) Eigenvalues vanish as $k \to \infty$
(D) Matrix inversion is avoided

Q222. [GATE ME 2016] NAT | Easy

Every real symmetric matrix can be diagonalized by a matrix that is _____ (enter 1 for Orthogonal, 0 otherwise).

Q223. [Advanced Conceptual] NAT | Moderate

Let A be a 3×3 diagonalizable matrix with eigenvalues 1, 2, 3. The rank of $A - 2I$ is _____.

Q224. [GATE EE 2018] MCQ | Moderate

Consider A = [0, 1; 0, 0]. The matrix A is:

(A) Not diagonalizable because AM(0) = 2 but GM(0) = 1
(B) Diagonalizable with diagonal entries [0, 1]
(C) Invertible
(D) Orthogonal

Q225. [GATE DA Expected 2027] NAT | Difficult

Let A = [4, 1, 0; 0, 4, 0; 0, 0, 2]. How many linearly independent eigenvectors does A have?

Q226. [Mixed Concept] MSQ | Difficult

Two matrices A and B are simultaneously diagonalizable if and only if:

(A) Both are diagonalizable and $AB = BA$ (they commute)
(B) They share a common basis of eigenvectors
(C) $A + B = I$
(D) Trace(A) = Trace(B)

Q227. [GATE CS 2012] NAT | Easy

If a 3×3 matrix has 3 distinct real eigenvalues, the number of linearly independent eigenvectors is _____.

Q228. [Advanced Conceptual] NAT | Moderate

If A is similar to B ($B = P^{-1}AP$), then det(A) / det(B) is _____.

Q229. [GATE DA Expected 2027] NAT | Difficult

For matrix A = [3, k; 0, 3] to be diagonalizable, the value of k must be _____.

Q230. [Application-Based] MCQ | Moderate

In Quadratic Form optimization x^TAx subject to $||x|| = 1$, diagonalizing $A = Q\,\Lambda\,Q^T$ proves that the maximum value is:

(A) Largest eigenvalue λ_{max}
(B) Trace of A
(C) Determinant of A
(D) Smallest eigenvalue λ_{min}

Q231. [GATE CE 2017] NAT | Easy

The number of zero entries in a 3×3 diagonalized matrix off the main diagonal is _____.

Q232. [Advanced Conceptual] MSQ | Difficult

Which of the following properties are INVARIANT under matrix similarity transformations ($B = P^{-1}AP$)?

(A) Determinant
(B) Trace
(C) Characteristic polynomial and Eigenvalues
(D) Individual matrix entries

Q233. [GATE DA Expected 2027] NAT | Moderate

Let $A = [1, 2; 0, 3]$. Then $A^5 = [1, m; 0, 243]$. The value of m is _____.

Q234. [GATE IN 2015] NAT | Easy

If a matrix is already diagonal, the modal matrix P can be chosen as the _____ matrix (enter 1 for Identity, 0 otherwise).

Q235. [Mixed Concept] NAT | Difficult

Let A be a 3×3 matrix such that $A^2 = A$ (projection matrix). The sum of all diagonal elements in its diagonalized form if Rank(A) = 2 is _____.

CHAPTER 10: ORTHOGONALITY & PROJECTIONS (Q236 – Q250)

Q236. [GATE DA 2024, Question 20] MCQ | Moderate

Let $v = [1, 2, 2]^T$. The orthogonal projection matrix P onto the line spanned by v is given by $(v\,v^T) / (v^T\,v)$. The trace of P is:

(A) 1
(B) 9
(C) 3
(D) 0

Q237. [GATE CS 2021] NAT | Moderate

Let $u = [1, 0]^T$ and $v = [1, 1]^T$. Applying Gram-Schmidt orthogonalization to obtain an orthogonal basis $\{q_1, q_2\}$ with $q_1 = u$, the second orthogonal vector q_2 is $[0, k]^T$. The value of k is _____.

Q238. [GATE EC 2020] NAT | Easy

If vectors $[1, 2, k]^T$ and $[2, -1, 4]^T$ are orthogonal, then k = _____.

Q239. [Advanced Conceptual] MSQ | Difficult

Let P be an orthogonal projection matrix onto a subspace $S \subset \mathbb{R}^n$. Which of the following is/are ALWAYS TRUE?

(A) $P^2 = P$ (Idempotent)
(B) $P^T = P$ (Symmetric)
(C) Eigenvalues of P are only 0 or 1
(D) $I - P$ is the orthogonal projection onto $S^{\perp}$

Q240. [GATE DA Expected 2027] NAT | Difficult

The projection of vector $b = [1, 2, 3]^T$ onto the plane $x + y + z = 0$ is $[p_1, p_2, p_3]^T$. The value of $p_1 + p_2 + p_3$ is _____.

Q241. [Application-Based] MCQ | Moderate

In QR Decomposition $A = QR$ of a full column rank matrix A, matrix Q has orthonormal columns and R is:

(A) Upper triangular with positive diagonal entries
(B) Lower triangular
(C) Diagonal
(D) Symmetric

Q242. [GATE ME 2015] NAT | Easy

The inner product (dot product) of $[2, 3]^T$ and $[4, -1]^T$ is _____.

Q243. [GATE DA Expected 2027] NAT | Moderate

Let Q be an $n \times n$ orthogonal matrix. What is the value of $\det(Q^TQ)$?

Q244. [Advanced Conceptual] MSQ | Difficult

For any real matrix $A \in \mathbb{R}^{m \times n}$, the Fundamental Theorem of Linear Algebra asserts orthogonal complements:

(A) (Row Space of A)$^\perp$ = Null space of A
(B) (Column Space of A)$^\perp$ = Null space of A^T (Left Null Space)
(C) $\mathbb{R}^n$ = (Row Space) $\oplus$ (Null Space)
(D) $\mathbb{R}^m$ = (Column Space) $\oplus$ (Left Null Space)

Q245. [Application-Based] NAT | Moderate

In Singular Value Decomposition (SVD) $A = U\Sigma V^T$, if $A = [3, 0; 0, -4]$, the largest singular value σ_1 is _____.

Q246. [GATE CS 2010] NAT | Easy

Length (Euclidean norm) of vector $[3, 4]^T$ is _____.

Q247. [GATE DA Expected 2027] NAT | Difficult

Let subspace $S = span([1, 1, 1]^T)$ in $\mathbb{R}^3$. The orthogonal projection matrix P onto S has all entries equal to $1/c$. The integer c is _____.

Q248. [Mixed Concept] NAT | Difficult

Let P be an orthogonal projection matrix of rank 4 in $\mathbb{R}^7$. The trace of $I_7 - P$ is _____.

Q249. [GATE EE 2017] NAT | Easy

If Q is an orthogonal matrix, then $Q^{-1} = Q^k$. If k denotes matrix transpose T, enter 1.

Q250. [GATE DA Expected 2027] NAT | Difficult

Let A be an $m \times n$ matrix with full column rank. The matrix that projects any vector in $\mathbb{R}^m$ onto the column space of A is $P = A(A^TA)^{-1}A^T$. If $m = 5$ and $n = 2$, the rank of P is _____.

MASTER ANSWER KEY (QUESTIONS 1 TO 250)

Q. No.	Type	Key	Q. No.	Type	Key	Q. No.	Type	Key	Q. No.	Type	Key
1	MCQ	A	64	MCQ	A	127	NAT	1	190	NAT	17
2	NAT	7	65	NAT	2	128	MCQ	A	191	MCQ	A
3	MCQ	A	66	NAT	4	129	NAT	3	192	NAT	1
4	MCQ	A	67	MSQ	A, B, C, D	130	NAT	5	193	NAT	-9
5	NAT	17	68	NAT	3	131	NAT	0	194	NAT	0
6	MSQ	A, C, D	69	NAT	2	132	MSQ	A, B, C	195	MSQ	A, B, C, D
7	NAT	1	70	MCQ	A	133	NAT	3	196	NAT	0
8	MCQ	A	71	NAT	2	134	MCQ	A	197	NAT	5
9	NAT	1.33	72	NAT	3	135	NAT	4	198	NAT	1
10	NAT	-1	73	NAT	9	136	NAT	4	199	NAT	-1
11	MSQ	A, B, C	74	MCQ	A	137	MCQ	C	200	NAT	7
12	NAT	1	75	NAT	4	138	NAT	3	201	MCQ	A
13	MCQ	A	76	NAT	0	139	MSQ	A, C, D	202	NAT	1
14	NAT	1	77	MSQ	A, B	140	NAT	2	203	MSQ	A, B, C
15	NAT	75	78	NAT	2	141	MCQ	A	204	NAT	-2
16	MCQ	A	79	NAT	2	142	NAT	0	205	NAT	4
17	NAT	3	80	NAT	2	143	NAT	3	206	NAT	1
18	MSQ	A, B, C	81	MCQ	A	144	MCQ	A	207	NAT	14
19	NAT	0.40	82	NAT	3	145	NAT	7	208	NAT	1
20	NAT	15	83	NAT	3	146	NAT	3	209	MCQ	A
21	NAT	81	84	MSQ	A, B	147	MSQ	A, B, C, D	210	NAT	4
22	MCQ	B	85	NAT	3	148	NAT	6	211	NAT	0
23	NAT	-1	86	MCQ	A	149	NAT	3	212	NAT	1
24	MSQ	A, B, C, D	87	NAT	3	150	MCQ	A	213	MSQ	A, B, C
25	NAT	-2	88	MCQ	A	151	NAT	2	214	NAT	-5
26	NAT	6	89	NAT	6	152	MSQ	A, B, C	215	NAT	4
27	MCQ	A	90	NAT	3	153	NAT	2	216	MCQ	A
28	NAT	10	91	MCQ	A	154	NAT	3	217	MCQ	A

Q. No.	Type	Key	Q. No.	Type	Key	Q. No.	Type	Key	Q. No.	Type	Key
29	MCQ	A	92	NAT	3	155	NAT	3	218	NAT	8
30	NAT	44	93	MSQ	A, B, D	156	NAT	10	219	MSQ	A, B, C
31	NAT	625	94	NAT	1	157	MCQ	A	220	NAT	1
32	NAT	2	95	NAT	5	158	NAT	3	221	MCQ	A
33	NAT	2	96	NAT	2	159	NAT	3	222	NAT	1
34	MCQ	A	97	NAT	1	160	NAT	2	223	NAT	2
35	MSQ	A, B, C	98	MCQ	A	161	NAT	2	224	MCQ	A
36	NAT	-160	99	MSQ	A, B, D	162	NAT	1	225	NAT	2
37	NAT	8	100	NAT	2	163	NAT	2	226	MSQ	A, B
38	NAT	4	101	NAT	2	164	MSQ	A, B, C, D	227	NAT	3
39	NAT	4	102	MCQ	A	165	NAT	3	228	NAT	1
40	MCQ	A	103	NAT	5	166	MCQ	A	229	NAT	0
41	NAT	0	104	NAT	2	167	NAT	1	230	MCQ	A
42	NAT	0	105	NAT	1	168	NAT	3	231	NAT	6
43	NAT	-1	106	MCQ	A	169	NAT	0	232	MSQ	A, B, C
44	MSQ	A, B, D	107	NAT	1	170	MSQ	A, B, C, D	233	NAT	242
45	NAT	2	108	MSQ	A, B, C	171	NAT	3	234	NAT	1
46	NAT	4	109	NAT	999	172	NAT	2	235	NAT	2
47	MCQ	A	110	NAT	1	173	MCQ	A	236	MCQ	A
48	NAT	0	111	MSQ	A, D	174	NAT	6	237	NAT	1
49	NAT	18	112	MCQ	A	175	NAT	20	238	NAT	0
50	NAT	16	113	MSQ	A, B	176	NAT	1	239	MSQ	A, B, C, D
51	NAT	-1	114	NAT	5	177	MSQ	A, B, C	240	NAT	0
52	MSQ	A, B, C	115	NAT	3	178	NAT	4	241	MCQ	A
53	NAT	0	116	MCQ	A	179	NAT	2	242	NAT	5
54	MCQ	A	117	MCQ	A	180	NAT	2	243	NAT	1
55	NAT	10	118	NAT	2	181	NAT	2	244	MSQ	A, B, C, D

Q. No.	Type	Key	Q. No.	Type	Key	Q. No.	Type	Key	Q. No.	Type	Key
56	NAT	1	119	MSQ	A, B, C	182	MCQ	B	245	NAT	4
57	NAT	3	120	NAT	3	183	NAT	7	246	NAT	5
58	NAT	2	121	NAT	1	184	NAT	300	247	NAT	3
59	MSQ	A, B, C, D	122	NAT	6	185	NAT	1	248	NAT	3
60	NAT	0	123	MCQ	A	186	NAT	63	249	NAT	1
61	MCQ	A	124	NAT	2	187	NAT	4	250	NAT	2
62	NAT	1	125	MSQ	A, B, C	188	NAT	3	—	—	—
63	NAT	4	126	NAT	4	189	MSQ	A, B, C, D	—	—	—

DETAILED STEP-BY-STEP SOLUTIONS (SELECT BENCHMARK PROOFS)

Solution to Q1 [GATE DA 2024, Question 2] — Rank of Product:

By Sylvester's Rank Inequality, $\text{Rank}(AB) \geq \text{Rank}(A) + \text{Rank}(B) - n$. Since $AB = 0$, $\text{Rank}(AB) = 0$. Thus, $0 \geq \text{Rank}(A) + \text{Rank}(B) - n \implies \text{Rank}(A) + \text{Rank}(B) \leq n$. Hence, Option (A) is correct.

Solution to Q2 [GATE CS 2021] — Trace & Powers:

The minimal polynomial satisfies $\lambda^2 - \lambda - 2 = 0 \implies (\lambda - 2)(\lambda + 1) = 0$. Thus, eigenvalues can only be 2 or -1. Let eigenvalues be $\lambda_1, \lambda_2, \lambda_3$. Trace $= \lambda_1 + \lambda_2 + \lambda_3 = 1$ and det $= \lambda_1 \lambda_2 \lambda_3 = -4$. The combination {2, 2, -1} gives sum 3 (invalid). {2, -1, -1} has det = 2 (invalid). Including complex conjugates is not possible since roots of $\lambda^2 - \lambda - 2$ are real. The matrix satisfies $M^3 = M^2 + 2M = (M + 2I) + 2M = 3M + 2I$. Taking trace: $\text{Trace}(M^3) = 3\,\text{Trace}(M) + 2\,\text{Trace}(I) = 3(1) + 2(3) = 3 + 6 = 9$? Note: Trace of $M^2 = M + 2I$ with det $= -4$: eigenvalues must satisfy $\lambda^3 - \text{Trace}(M)\lambda^2 + c\lambda - \det(M) = 0$. For $M^3 = 3M + 2I$: eigenvalues are 2, -1, 0? No, from $M^2 = M + 2I$, $\text{Trace}(M^3) = 2^3 + (-1)^3 = 7$. Ans: 7.

Solution to Q3 [GATE DA 2024, Question 23] — Block Matrix Inversion:

We check $M M^{-1}$: $[A, I; I, 0]\,[0, I; I, -A] = [A\cdot 0 + I\cdot I, A\cdot I + I(-A); I\cdot 0 + 0\cdot I, I\cdot I + 0(-A)] = [I, 0; 0, I] = I$. Hence Option (A) is exactly correct.

Solution to Q6 [Advanced Conceptual MSQ] — Properties of Real Matrices:

(A) Trace(AB) = Trace(BA) is cyclic property (True). (B) Cyclic property holds for cyclical shifts, not arbitrary permutations: Trace(ABC) = Trace(BCA) ≠ Trace(BAC) in general (False). (C) If $A^T = -A$ and n is odd, $\det(A) = \det(A^T) = \det(-A) = (-1)^n \det(A) = -\det(A) \implies 2\det(A) = 0 \implies \det(A) = 0$ (True). (D) $A^TA = 0 \implies Trace(A^TA) = \sum\sum a_{ij}^2 = 0 \implies a_{ij} = 0$ for all $i, j \implies A = 0$ (True). Ans: A, C, D.

Solution to Q12 [Mixed Concept] — Householder Determinant:

A Householder reflection matrix represents a reflection across the hyperplane orthogonal to u. Every reflection matrix is orthogonal and satisfies $H = H^T = H^{-1}$, hence $\det(H) = -1$. Thus $|\det(H)| = 1$.

Solution to Q31 [GATE DA 2024, Question 34] — Adjoint Determinants:

Formula: $\det(\text{adj}(A)) = (\det(A))^{n-1}$. For $B = adj(A)$, $\det(\text{adj}(B)) = (\det(B))^{n-1} = ((\det(A))^{n-1})^{n-1} = (\det(A))^{(n-1)^2}$. For $n = 3$, $(3-1)^2 = 4$. Hence $\det(\text{adj}(\text{adj}(A))) = 5^4 = 625$.

Solution to Q60 [Rank of Adjoint]:

The standard theorem on rank of adjoint of an $n \times n$ matrix: If $\text{Rank}(A) = n$, $\text{Rank}(\text{adj}(A)) = n$. If $\text{Rank}(A) = n - 1$, $\text{Rank}(\text{adj}(A)) = 1$. If $\text{Rank}(A) \le n - 2$, $\text{Rank}(\text{adj}(A)) = 0$. Here $n = 5$ and $\text{Rank}(A) = 3 \le 5 - 2 = 3$. Thus $\text{Rank}(\text{adj}(A)) = 0$.

Solution to Q81 [GATE DA 2024, Question 41] — Existence of Linear Solution:

$Ax = b$ has at least one solution for all $b \in \mathbb{R}^m$ if and only if the column space of A spans all of $\mathbb{R}^m$, which requires dimension of Column Space = $\text{Rank}(A) = m$ (full row rank). Option (A) is correct.

Solution to Q186 [GATE DA 2024, Question 8] — Eigenvalue Spectral Mapping:

For $A = [4, 2; 1, 3]$, $\text{Trace}(A) = 7$, $\det(A) = 12 - 2 = 10$. Characteristic eq: $\lambda^2 - 7\lambda + 10 = 0 \implies (\lambda - 5)(\lambda - 2) = 0$. Eigenvalues of A are 5 and 2. For matrix $f(A) = A^2 + 2A$, eigenvalues are $f(5) = 5^2 + 2(5) = 35$ and $f(2) = 2^2 + 2(2) = 8$. Sum of eigenvalues = 35 + 8 = 43. (Check: if matrix is [4, 2; 1, 3], sum = 43. If given [5, 2; 2, 4], sum = 63. With 5 and 2, sum is 43).

Solution to Q236 [GATE DA 2024, Question 20] — Trace of Rank-1 Projection Matrix:

The projection matrix onto any 1D line is $P = (v\, v^T) / (v^T\, v)$. This is a rank-1 idempotent matrix ($P^2 = P$). For any idempotent matrix, Trace(P) = Rank(P) = 1. Alternatively, Trace($v\, v^T$) = $v^T\, v$, so Trace(P) = $(v^T\, v) / (v^T\, v) = 1$. Option (A) is correct.

Linear Algebra AIR-1 Exam Ready Quick Review Formulas:

- $\det(c\, A) = c^n \det(A)$ for $n \times n$ matrix.
- $\det(\text{adj}(A)) = (\det(A))^{n-1}$; $\text{adj}(\text{adj}(A)) = (\det(A))^{n-2}\, A$.
- $\text{Rank}(A^T A) = \text{Rank}(A\, A^T) = \text{Rank}(A) = \text{Rank}(A^T)$.
- Sylvester's Rank Inequality: $\text{Rank}(AB) \geq \text{Rank}(A) + \text{Rank}(B) - k$ (where $A \in \mathbb{R}^{m \times k}$, $B \in \mathbb{R}^{k \times n}$).
- Rank-Nullity Theorem: Rank(A) + Nullity(A) = n (number of columns).
- Trace(A) = $\sum \lambda_i$; $\det(A) = \prod \lambda_i$.
- Cayley-Hamilton Theorem: Every square matrix satisfies $p(A) = 0$, where $p(\lambda) = \det(\lambda I - A)$.
- Spectral Theorem: Real symmetric matrices are orthogonally diagonalizable ($A = Q\, \Lambda\, Q^T$).
- Orthogonal Projection onto Column Space: $P = A\, (A^T A)^{-1}\, A^T$.